

Keshav Krishna

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Summary

Software engineer with two years of experience in data science, machine learning, and backend development. Designed and optimized high-throughput content recommendation services serving over 12 million daily users and developed custom transformer and few-shot learning models to enhance content moderation accuracy. Proficient in Python, Spark, Hive, Flask, Docker, and deep learning frameworks. Seeking to apply generative AI expertise and distributed system fundamentals to support scalable platform solutions.

EDUCATION

New York University Courant

MS, Computer Science (GPA: 4.0)

- **Coursework:** Machine Learning, Honors Analysis of Algorithms, Programming Languages

Aug 2025 - May 2027

New York City

Indian Institute of Technology, Ropar

Bachelors, Computer Science (GPA: 8.74/10)

- **Achievements:** Merit Scholarship for top performance

Aug 2019 - Jun 2023

Ropar, India

SKILLS SUMMARY

- **Programming Languages:** C, C++, Python, SQL, Unix scripting, Java
- **Machine Learning & AI:** Machine learning, Deep learning, LLMs, Tensorflow, Hugging Face, Pytorch, Generative AI, Computer Vision, Image Processing, Video Processing
- **Data Engineering & Tools:** Docker, GIT, JIRA, Postgres, redis, FastAPI, MYSQL, Airflow, pyspark, hive, Qdrant, Annoy, Hadoop, NoSQL, ETL, databases, JSON
- **Software Engineering & Architecture:** Backend, Agile, Software Engineering Fundamentals, Data Structures, Distributed System Design

EXPERIENCE

JioSaavn | Software Engineer(Data Science)

Jul 2023 - Jul 2025

- Built distributed backend services for early-morning and late-night content mixes for 12M daily users, employing software engineering fundamentals and distributed system design with Hive, Spark, Redis, and PHP, driving 1M impressions, 800K clicks, and 500K streams per day
- Generated mood-based topic mixes for 7.6M daily users using Hive, Spark, and TF-IDF, added deduplication filters and Flask endpoints to the PHP backend, and served them to 100M app users, achieving a 12.81% CTR and 8.5 streams per user
- Assessed few-shot SetFit and transformer models (hing-roberta, muril-base) and transitioned to prompt engineering with Llama3-8B-instruct, leveraging generative AI techniques to achieve 85% accuracy on manually labeled data, improving content moderation of user-generated playlists
- Created a custom BERT model to generate song and user embeddings, which raised top-5 and top-10 accuracy to 61.82% and 67.21%, thereby improving user-learning capabilities and recommendation personalization
- Developed a RAG system using LLaMA3-7B for prompt intent classification, dynamically retrieving songs using optimized data structures to assemble playlists and generating custom cover art via the segmid/SSD-1B diffusion model, enhancing user interaction and playlist customization
- Reduced infrastructure costs by migrating data from Annoy indices to Qdrant, purging unused data, and cleaning tables for improved lifecycle management

GE Healthcare | Software Intern

May 2022 - Jul 2022

- Applied YOLO algorithm in a computer vision pipeline to train a deep learning model that identified and blurred faces in medical images, achieving over 90% accuracy on more than 20,000 images
- Trained YOLO models for people detection in IR and RGB images, applying video processing and image processing techniques to reach over 85% mean average precision and accuracy
- Created Docker images with globally accessible modules, simplifying import and utilization in various projects by implementing FastAPI endpoints

RESEARCH & ACADEMIC PROJECTS

Review of machine learning innovations in cache management

Jan 2024 - Apr 2024

<https://www.frontiersin.org/journals/artificial-intelligence/articles/10.3389/frai.2025.1441250/abstract>

Independent Research

- Single-author literature review of 40+ papers on ML for cache systems (replacement policies, dynamic management, edge/content caching, security).
- Evaluated algorithmic approaches including reinforcement learning, LSTM, imitation learning, and deep neural networks for performance optimization and attack detection in cache systems
- Published in Frontiers in AI; provided roadmap for research and practitioner adoption with 9+ citations in few months.

FlashAttention-2: From-Scratch Triton Implementation with GPU Profiling & Benchmarking

Jan 2026 - Mar 2026

<https://github.com/k-keshav-k/Flash-Attention>

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- Implemented FlashAttention-2 forward pass using Triton (C++) and Python, with online softmax and tiled computation, reducing attention memory from $O(N^2)$ to $O(N)$ for sequences up to 65K
- Built backward pass with activation recomputation from stored logsumexp values and integrated with PyTorch's torch.compile for efficient kernel fusion across FP32 and BF16
- Profiled Transformer models up to 2.7B parameters with NVIDIA Nsight Systems and PyTorch memory snapshots; benchmarked Triton kernel against vanilla PyTorch and baselines across sequence lengths, dimensions, and precisions

Pico-LLM: Unified SLM Training & Interpretability | <https://github.com/k-keshav-k/pico-llm/tree/main>

Sep 2025 - Dec 2025

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- **Unified Training Framework:** Developed a codebase for Small Language Models (SLMs) training from scratch along with comparison with lstm and kmeans.
- **Mathematical Reasoning:** Extended base model trained to also include reasoning by supporting **SFT** and **DPO** alignment on reasoning datasets like GSM8K.
- **Mechanistic Interpretability:** Implemented visualization tools (Attention Maps, Training loss, Test loss, T-SNE) to diagnose feature activations and internal model states.
- **Inference Optimization:** Engineered a robust inference engine with **KV-Caching** and automatic architecture detection for efficient batched generation.